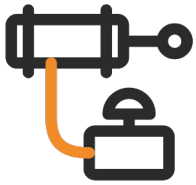
Name: _____ Class: _____

N5 Engineering Science

Booklet 6: Pneumatics

Lessons Pn1 to Pn8



My progress

Lesson	Learning intention	✓
Pn1	**Pneumatic Systems and Air Types** To describe a pneumatic system and name the air types on QualScot circuits.	☐
Pn2	**The Single-Acting Cylinder and the 3/2 Valve** To describe how a 3/2 valve controls a single-acting cylinder, in QualScot words.	☐
Pn3	**The Double-Acting Cylinder, the 5/2 Valve and Pilot Air** To describe how pilot air and a 5/2 valve control a double-acting cylinder.	☐
Pn4	**Controlling Speed: Restrictors** To describe how a restrictor and a uni-directional restrictor control the speed of a piston.	☐
Pn5	**Time Delays: the Reservoir** To describe how a uni-directional restrictor and a reservoir make a time delay.	☐
Pn6	**AND and OR Control** To pipe and describe AND control (valves in series) and OR control (a shuttle valve).	☐
Pn7	**Force, Pressure and Area** To calculate the force, pressure or area for a pneumatic cylinder using F = PA.	☐
Pn8	**Describing and Designing Circuits** To describe a circuit in QualScot words and design one to a specification.	☐

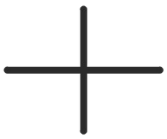
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Data Sheet: Pneumatic Symbols

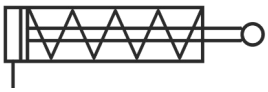
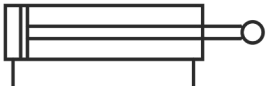
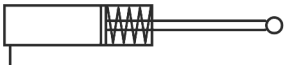
Air

 Main air	Pilot air	 Exhaust	

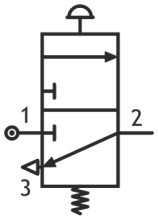
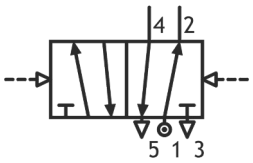
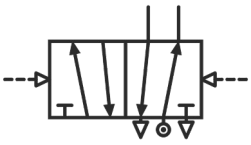
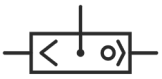
Pipes

Main air pipe	Pilot air pipe	 Joined	 Crossing

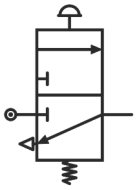
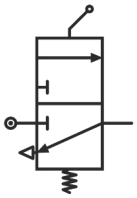
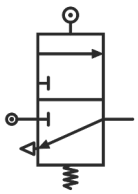
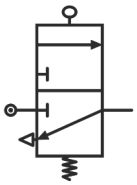
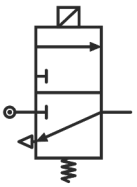
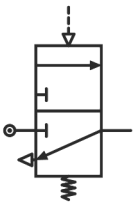
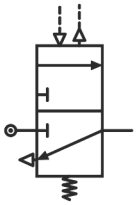
Cylinders

 Single-acting	 Double-acting	 Single-acting, out	

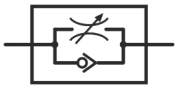
Valves

 3/2 valve	 5/2 valve	 5/2 valve, pilot/pilot	 Shuttle valve

Actuators (on a 3/2 valve)

 Push button	 Lever	 Roller	 Plunger	 Solenoid	 Pilot air	 Diaphragm

Speed and time

 Restrictor	 Uni-directional restrictor	 Reservoir	 Air bleed

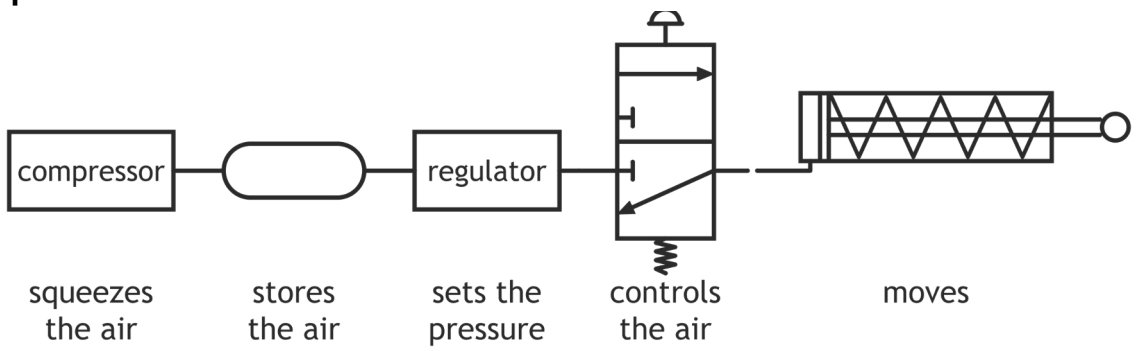
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Lesson Pn1: Pneumatic Systems and Air Types

LI: To describe a pneumatic system and name the air types on QualScot circuits.

SC:
- Pn1.1 I can **describe** what pneumatics is and **give** uses of it. ☐
- Pn1.2 I can **name** the three air types: main air, pilot air and exhaust. ☐
- Pn1.3 I can **describe** the function of main air and of pilot air. ☐
- Pn1.4 I can **read** the pipes on a circuit: main air, pilot air, joined and crossing. ☐
- Pn1.5 I can **follow** the safety rules for the pneumatics kit. ☐

What is pneumatics?



Pneumatics uses **compressed** air to make things move.
A **compressor** squeezes air and stores it in a **reservoir**.
A regulator sets the working **pressure** before the air reaches the circuit.
Pneumatics is used on **production** lines to push, clamp, lift and stamp products.
Bus and train doors and fairground rides also use **pneumatic** cylinders.

Air types

Symbol	Name	What it does
	**main air**	Main air operates a **cylinder**: it makes the piston outstroke or instroke.
	pilot air	Pilot air is a small air signal used to **actuate** a valve.
	**exhaust**	Exhaust is where used air leaves the circuit to the **atmosphere**.

Quick check

1 **State** where used air leaves the circuit.

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Pipes

Symbol	Name	What it means
———	**main air pipe**	A **solid** line carries main air.
- - - - - - -	**pilot air pipe**	A **dashed** line carries pilot air.
	**joined**	A **dot** shows that the pipes are joined.
	**crossing**	Pipes that cross without a dot are **separate** pipes.

Safety with the kit

Wear **goggles** whenever the air is on.
Turn the air **off** before you connect or take apart any pipe.
Push each pipe fully **in** and pull gently to check it is held.
Never point a pipe or an exhaust at **anyone**.
Keep fingers clear of a **moving** piston.

Check your understanding

1 **State** what pneumatics uses to make things move.

2 **State** the function of **main air** and of **pilot air**.

3 **State** how a pilot air pipe is drawn, and how a main air pipe is drawn.

4 **State** two safety rules for the pneumatics kit.

Review Questions: now complete the Lesson Pn1 questions in your jotter.
Scan the QR code or go to **mrstewartphysics.co.uk/rq/pn1** · Tap **Show answer** to mark each one.

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Lesson Pn2: The Single-Acting Cylinder and the 3/2 Valve

LI: To describe how a 3/2 valve controls a single-acting cylinder, in QualScot words.

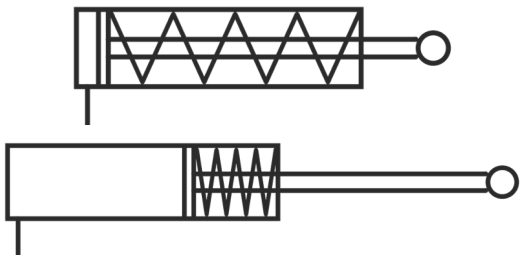
SC: Pn2.1 I can **describe** how a single-acting cylinder outstrokes and instrokes. ☐

Pn2.2 I can **read** a 3/2 valve symbol: its ports, its two states, main air and exhaust. ☐

Pn2.3 I can **name** the ways a valve is actuated, from its symbol. ☐

Pn2.4 I can **describe** a 3/2 valve circuit using actuated, outstrokes and instrokes. ☐

The single-acting cylinder

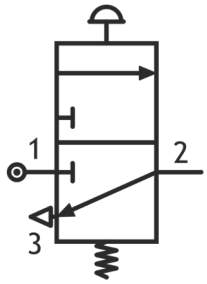


Main air pushes the piston, so the piston **outstrokes**.

When the air is exhausted, the **spring** pushes the piston back: it **instrokes**.

A single-acting cylinder is powered by air in **one** direction only.

The 3/2 valve



3/2 means **3** ports and **2** states.

Port 1 is **main air**, port 2 goes to the **cylinder** and port 3 is **exhaust**.

Each box shows one state. The box beside the spring is the **unactuated** state.

Actuated: air flows from port 1 to port **2**, so the piston outstrokes.

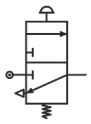
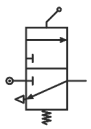
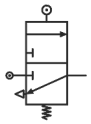
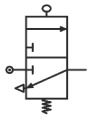
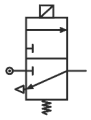
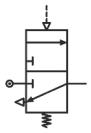
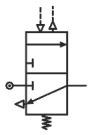
Unactuated: air flows from port 2 to port **3**, so the air exhausts and the piston instrokes.

Quick check

1 **State** what returns a 3/2 valve to its unactuated state.

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Ways to actuate a valve

Symbol	Actuator	What operates it
	**push button**	Operated by a person **presses** it.
	**lever**	Operated by a person **moves** it.
	**roller**	Operated by a moving part **rolls** over it.
	**plunger**	Operated by a moving part **pushes** it.
	**solenoid**	Operated by an **electrical** signal, e.g. from a microcontroller.
	**pilot air**	Operated by an **air** signal from another valve.
	**diaphragm**	Operated by a change in air **pressure**, e.g. an air bleed being covered.

The spring under the valve returns it to the **unactuated** state when the actuator is released.

Quick check

1 **Choose** an actuator for a valve that the moving piston operates.

Choosing an actuator

Choose the actuator from what must operate the valve in the task.

A person starts it: **push button** or lever. A moving part must operate it: **roller** or plunger.

An electrical signal, e.g. from a microcontroller or sensor: **solenoid**. An object must be sensed without touching it: **diaphragm** with an air bleed.

Justify it **in context**: "A roller, because the moving piston rolls over it at the end of its outstroke."

Name a valve in full, e.g. "3/2 push-button spring-return valve".

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Words that get the marks

When you describe a circuit, use:

- the valve is **actuated** (or unactuated)
- the piston **outstrokes** or **instrokes**
- name every valve and piston by its number or letter

Model answer

When the push button is pressed, valve 1 is **actuated**. Main air flows to the cylinder, so the piston **outstrokes**. When the button is released, the spring returns valve 1 and the air **exhausts**. The spring in the cylinder makes the piston **instroke**.

Check your understanding

1 **State** what 3/2 means.

2 **State** what makes the piston of a single-acting cylinder instroke.

3 **Name** the ports of a 3/2 valve: port 1, port 2 and port 3.

4 **Name** the actuator that is operated by an electrical signal.

5 **Describe** what happens when the push button on a 3/2 valve piped to a single-acting cylinder is pressed.

Review Questions: now complete the Lesson Pn2 questions in your jotter.
Scan the QR code or go to **mrstewartphysics.co.uk/rq/pn2** · Tap **Show answer** to mark each one.

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Build It: the Push-Button Clamp

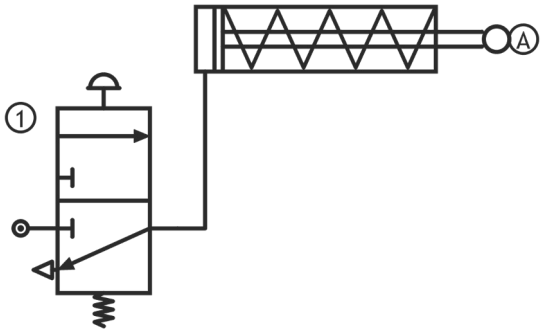
A clamp holds a part still while it is drilled. A pneumatic circuit outstrokes a piston when a push button on a valve is pressed. This task is from 2024 Section 1 Question 3.

The following specification has been written:

A The piston outstrokes when the push button on valve 1 is pressed.

B The piston instrokes when the push button is released.

C The system must be repeatable.



Pn2a Write a test plan: three tests and the expected result of each. After you have simulated or built it, record each test result. Say what the piston did.

Planned test	Expected result	Test result
Press the push button on valve 1.		
Release the push button.		
Repeat tests 1 and 2 three times.		

Pn2b Simulate the circuit, then construct it on the kit.

1. Open NoPressureSim (QR code, or sites.google.com/view/nostrainsim/pneumatics). **Build** the circuit. Show the main air supply on every valve in your evidence.

2. Predict what happens. Test it. Was your prediction right?

3. Air off. **Build** it on the kit: a 3/2 push-button spring-return valve and a single-acting cylinder.

4. Air on. Carry out your three tests. **Record** each test result.

Explain with cause and effect. Say what you did and what the piston did. Point A is done for you.

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Pn2c Evaluate your solution. For each specification point, state if it was met and explain why, using your test results.

Specification point	Met? Yes/No	Explanation
A	Yes	When the push button on valve 1 was pressed, valve 1 was actuated and piston A outstroked.
B		
C		

Pn2d The operator's hand is close to the clamp when they press the button. **Describe** a modification that would solve this. Justify it.

Modification	
Justification	

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Lesson Pn3: The Double-Acting Cylinder, the 5/2 Valve and Pilot Air

LI: To describe how pilot air and a 5/2 valve control a double-acting cylinder.

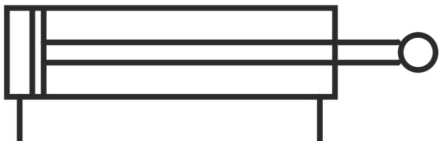
SC: Pn3.1 I can **describe** how a double-acting cylinder outstrokes and instrokes. ☐

Pn3.2 I can **read** a 5/2 valve symbol: its ports, its two states, main air and exhausts. ☐

Pn3.3 I can **describe** how pilot air changes the state of a 5/2 valve. ☐

Pn3.4 I can **describe** a circuit that outstrokes and instrokes a piston automatically. ☐

The double-acting cylinder

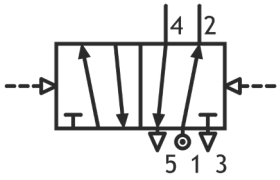


Main air on the left of the piston makes it **outstroke**.

Main air on the right of the piston makes it **instroke**.

A double-acting cylinder has no spring: it is powered by air in **both** directions.

The 5/2 valve



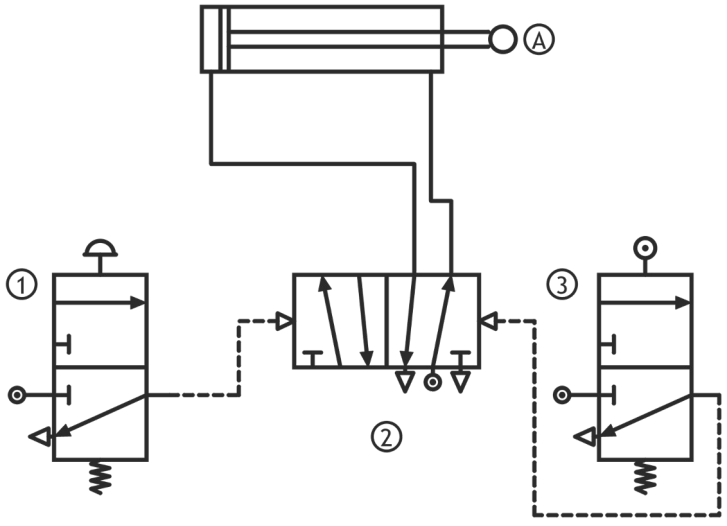
5/2 means **5** ports and **2** states.

Port 1 is main air. Ports 2 and 4 go to the **cylinder**. Ports 3 and 5 are **exhausts**.

When pilot air reaches one end, the 5/2 valve **changes state**.

The valve stays in that state until pilot air reaches the **other** end.

Reciprocating motion



Valve 1 is actuated. Pilot air changes the state of valve 2, so the piston **outstrokes**.

At the end of its outstroke the piston **actuates** valve 3.

Pilot air from valve 3 changes the state of valve 2 back, so the piston **instrokes**.

Outstroke then instroke, over and over, is **reciprocating** motion.

Quick check

1 **State** which ports of a 5/2 valve go to the cylinder.

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Each mark needs three things: the **valve** by its number, what it does, and what the **named** piston does.

To complete the piping, join ports 4 and 2 of the 5/2 valve to **both** ends of the cylinder. Each pipe goes port to port.

Check your understanding

1 **State** what 5/2 means.

2 **State** the difference between a single-acting and a double-acting cylinder.

3 **State** what makes a 5/2 pilot/pilot valve change state.

4 Pilot air stops reaching a 5/2 pilot/pilot valve. **State** what the valve does.

5 **Name** the motion when a piston outstrokes then instrokes, over and over.

6 **State** what actuates the roller valve at the end of the outstroke.

Review Questions: now complete the Lesson Pn3 questions in your jotter.
Scan the QR code or go to **mrstewartphysics.co.uk/rq/pn3** · Tap **Show answer** to mark each one.

Build It: the Reciprocating Saw

A pneumatic piston moves a saw blade back and forth. The piston must outstroke, then instroke automatically when it reaches the end of its outstroke.

The following specification has been written:

A Piston A outstrokes when the push button on valve 1 is pressed.

B Piston A instrokes when it actuates roller valve 3 at the end of its outstroke.

C The system must be repeatable.

The circuit is shown in the notes above.

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Pn3a Write a test plan: three tests and the expected result of each. After you have simulated or built it, record each test result. Say what the piston did.

Planned test	Expected result	Test result
Press the push button on valve 1.	Piston A outstrokes.	

Pn3b Simulate the circuit, then construct it on the kit.

1. Open NoPressureSim (QR code). **Build** the circuit, as on page 8. Show the main air supply on every valve.

2. Predict what happens. Test it. Was your prediction right?

3. Air off. **Build** it on the kit: push-button valve 1, 5/2 valve 2, roller valve 3 and a double-acting cylinder.

4. Air on. Carry out your three tests. **Record** each test result.

Pn3c Evaluate your solution. For each specification point, state if it was met and explain why, using your test results.

Specification point	Met? Yes/No	Explanation
A		
B		
C		

Pn3d The operator has to stand next to the saw blade to press the button. **Describe** a modification that would solve this. Justify it.

Modification	
Justification	

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Lesson Pn4: Controlling Speed: Restrictors

LI: To describe how a restrictor and a uni-directional restrictor control the speed of a piston.

SC: Pn4.1 I can **identify** a restrictor and a uni-directional restrictor from their symbols. ☐

Pn4.2 I can **describe** how each one controls the speed of a piston. ☐

Pn4.3 I can **describe** where a uni-directional restrictor goes to slow only one stroke. ☐

Pn4.4 I can **draw** a uni-directional restrictor the right way round for the air flow. ☐

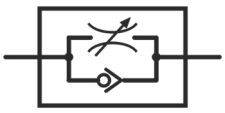
The restrictor



A restrictor makes the path for the air **narrower**, so air flows more slowly in **both** directions.

The arrow through it shows the restriction can be **adjusted**.

The uni-directional restrictor



A uni-directional restrictor slows the air in **one** direction only.

In the other direction the air flows freely through the **non-return** valve (the by-pass).

Where to fit it

To slow a piston **smoothly**, restrict the air **leaving** the cylinder (the exhaust air).

To slow the outstroke only, fit it in the pipe to the **instroking** side of the cylinder.

Say **how**: "add a uni-directional restrictor to slow the exhaust air". "UDR" on its own gets no mark.

Check your understanding

1 **State** the difference between a restrictor and a uni-directional restrictor.

2 **Draw** the symbol for a uni-directional restrictor.

Space for working and answer

3 **State** what the arrow on a restrictor symbol shows.

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4 **State** which air should be restricted to slow a piston smoothly.

---

5 **Explain** why "fit a UDR" gets no mark.

Review Questions: now complete the Lesson Pn4 questions in your jotter.
Scan the QR code or go to **mrstewartphysics.co.uk/rq/pn4** · Tap **Show answer** to mark each one.

Build It: the Slow Door

A pneumatic piston opens a door. The door must open slowly and smoothly so that it is safe (2024 Section 2 Question 15).

The following specification has been written:

A Piston A outstrokes when valve 1 is actuated.

B Piston A outstrokes slowly and smoothly.

C The speed of the instroke is not changed.

Pn4a Start from the Lesson Pn3 circuit and add a uni-directional restrictor. **Draw** the circuit in the box, using the symbols on the data sheet (page 2). Label every valve with a number.

Pn4b Write a test plan: three tests and the expected result of each. After you have simulated or built it, record each test result. Say what the piston did.

Planned test	Expected result	Test result

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Planned test	Expected result	Test result

Pn4c Simulate the circuit, then construct it on the kit.

1. Open NoPressureSim (QR code). **Build** the circuit, as on page 8. Show the main air supply on every valve.

2. Predict what happens. Test it. Was your prediction right?

3. Air off. **Build** your circuit on the kit.

4. Air on. Carry out your three tests. **Record** each test result.

Pn4d Evaluate your solution. For each specification point, state if it was met and explain why, using your test results.

Specification point	Met? Yes/No	Explanation
A		
B		
C		

Pn4e The door now opens safely, but it closes too fast. **Describe** a modification that would solve this. Justify it.

Modification	
Justification	

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Lesson Pn5: Time Delays: the Reservoir

LI: To describe how a uni-directional restrictor and a reservoir make a time delay.

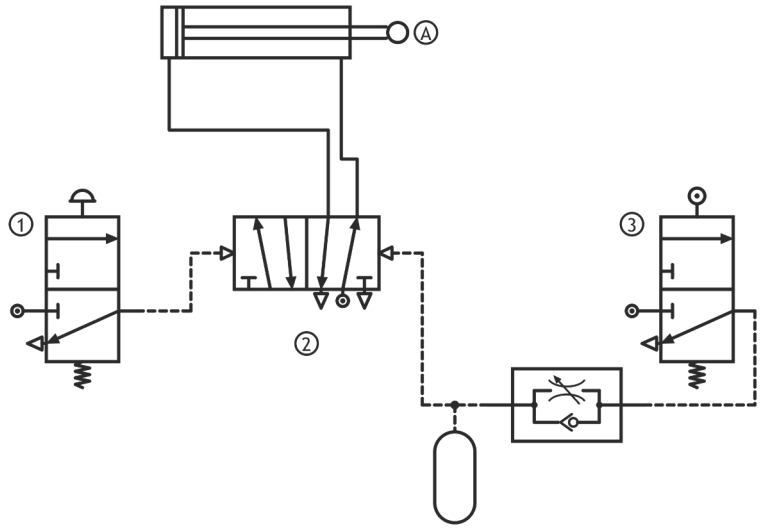
SC: Pn5.1 I can **describe** how a uni-directional restrictor and a reservoir make a time delay. ☐

Pn5.2 I can **draw** a uni-directional restrictor and reservoir the right way round. ☐

Pn5.3 I can **describe** how to make a time delay longer or shorter. ☐

Pn5.4 I can **describe** the operation of a circuit that has a time delay. ☐

How a time delay works



A **reservoir** is an empty tank. Air must fill it before the pressure in it builds up.

A uni-directional restrictor slows the air going **into** the reservoir, so the reservoir fills slowly.

When the pressure is high enough, the pilot air **changes the state** of the valve.

The time the reservoir takes to fill is the **time delay**.

The arrow on the restrictor means the time delay is **adjustable**.

Changing the delay

To make the delay **longer**: close the restrictor more, or use a **bigger** reservoir.
To make the delay **shorter**: open the restrictor more, or use a smaller reservoir.

A drawn time delay gets a mark for each of these. The uni-directional restrictor **symbol**. The reservoir symbol. The restrictor in the right **position**, the right way round: it slows the air going into the reservoir.

Check your understanding

1 **State** what a reservoir is.

2 **State** where the uni-directional restrictor goes in a time delay.

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3 **Draw** a uni-directional restrictor and a reservoir to make a time delay. Air flows from left to right.

Space for working and answer

4 **State** two ways to make a time delay longer.

5 **Explain** why a bigger reservoir gives a longer time delay.

Review Questions: now complete the Lesson Pn5 questions in your jotter.
Scan the QR code or go to **mrstewartphysics.co.uk/rq/pn5** · Tap **Show answer** to mark each one.

Build It: the Glue Press

A press holds two parts together while the glue sets, then lets them go.

The following specification has been written:

A Piston A outstrokes when valve 1 is actuated.
B Piston A stays outstroked for an adjustable time after it actuates valve 3.
C Piston A then instrokes automatically.

The circuit is shown in the notes above.

Pn5a **Write** a test plan: three tests and the expected result of each. After you have simulated or built it, record each test result. Say what the piston did.

Planned test	Expected result	Test result

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Pn5b Simulate the circuit, then construct it on the kit.

1. Open NoPressureSim (QR code). **Build** the circuit, as on page 8. Show the main air supply on every valve.

2. Predict what happens. Test it. Was your prediction right?

3. Air off. **Build** the Lesson Pn3 circuit. **Add** the uni-directional restrictor and the reservoir.

4. Air on. Carry out your three tests. **Record** each test result.

Pn5c Evaluate your solution. For each specification point, state if it was met and explain why, using your test results.

Specification point	Met? Yes/No	Explanation
A		
B		
C		

Pn5d Some glues need a much longer time to set. **Describe** a modification that would solve this. Justify it.

Modification	
Justification	

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Lesson Pn6: AND and OR Control

LI: To pipe and describe AND control (valves in series) and OR control (a shuttle valve).

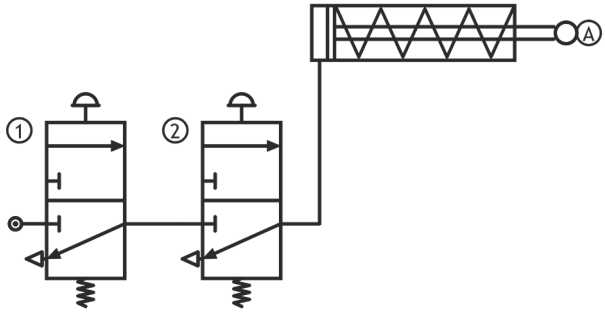
SC: Pn6.1 I can **pipe** two valves in series to give AND control. ☐

Pn6.2 I can **pipe** two valves and a shuttle valve to give OR control. ☐

Pn6.3 I can **state** the type of control a circuit uses, from its piping. ☐

Pn6.4 I can **explain** why two-hand AND control keeps an operator safe. ☐

AND control

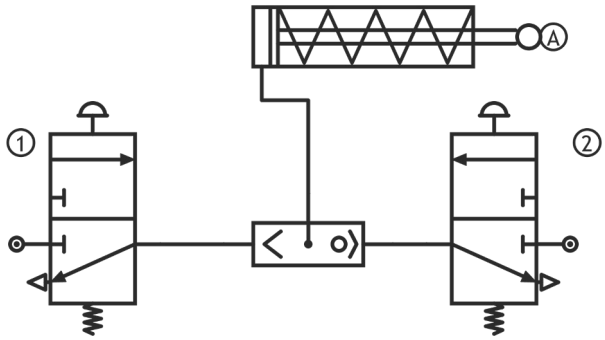


AND control: the piston moves only when valve 1 **and** valve 2 are both actuated.

The valves are piped in **series**: the output of valve 1 is the main air supply for valve 2.

Two-hand control is AND control. Both hands must be on the buttons, so they are kept **away** from the piston.

OR control



OR control: the piston moves when valve 1 **or** valve 2 is actuated.

The two valves are joined by a **shuttle valve**.

Air from either valve pushes the ball across. The air goes to the output, not out through the other valve's **exhaust**.

Do not join the two pipes with a T: that is not OR control, and it gets no mark.

Pipe the output of the AND valves to the **pilot** end of the 5/2 valve. Do the same with the output of a shuttle valve. Draw that pipe dashed: it is pilot air.

Check your understanding

1 **State** the type of control where two valves must both be actuated.

---

2 **State** how the two valves are piped for AND control.

---

3 **Name** the valve used to join two valves for OR control.

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4 **State** why two pipes must not be joined with a T for OR control.

5 **Explain** why two-hand control keeps an operator safe.

Review Questions: now complete the Lesson Pn6 questions in your jotter.
Scan the QR code or go to **mrstewartphysics.co.uk/rq/pn6** · Tap **Show answer** to mark each one.

Build It: the Two-Hand Press

A press stamps a shape into a metal sheet. For safety, both of the operator's hands must be on the buttons before the press moves.

The following specification has been written:

A Piston A outstrokes only when valve 1 and valve 2 are both actuated.

B Piston A does not outstroke when only one valve is actuated.

C Piston A instrokes automatically at the end of its outstroke.

Pn6a Draw the circuit in the box, using the symbols on the data sheet (page 2). Label every valve with a number.

Pn6b Choose the actuator for valve 4, the valve that makes the piston instroke. **Justify** your choice.

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Pn6c Write a test plan: three tests and the expected result of each. After you have simulated or built it, record each test result. Say what the piston did.

Planned test	Expected result	Test result

Pn6d Simulate the circuit, then construct it on the kit.

1. Open NoPressureSim (QR code). **Build** the circuit, as on page 8. Show the main air supply on every valve.
2. Predict what happens. Test it. Was your prediction right?
3. Air off. **Build** your circuit on the kit.
4. Air on. Carry out your three tests. **Record** each test result.

Pn6e Evaluate your solution. For each specification point, state if it was met and explain why, using your test results.

Specification point	Met? Yes/No	Explanation
A		
B		
C		

Pn6f The press comes down very fast and slams into the metal sheet. **Describe** a modification that would solve this. Justify it.

Modification	
Justification	

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Lesson Pn7: Force, Pressure and Area

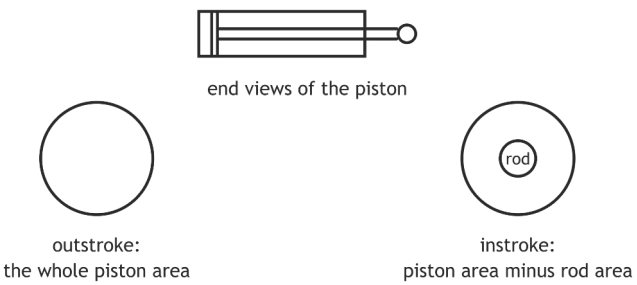
LI: To calculate the force, pressure or area for a pneumatic cylinder using F = PA.

SC:
- Pn7.1 I can **calculate** the area of a piston from its diameter. ☐
- Pn7.2 I can **calculate** the outstroke force of a cylinder using F = PA. ☐
- Pn7.3 I can **calculate** a pressure or an area by rearranging F = PA. ☐
- Pn7.4 I can **calculate** the instroke force using the effective area. ☐
- Pn7.5 I can **explain** why the instroke force is smaller than the outstroke force. ☐

The formula

Pressure is force per unit **area**: P = F ÷ A, so **F = P × A**.
F = force in newtons (**N**). P = pressure in **N mm^{-2}**. A = area in **mm^2**.
The area of a piston is **A = πr^2**, where r is the radius: half the diameter. (Or A = πd^2 ÷ 4.)

Outstroke and instroke



Outstroke: air pushes on the **whole** piston area.

Instroke: the rod takes up part of the piston, so the air pushes on a smaller area.

Effective area = piston area − rod area.

With the same pressure, the instroke force is **smaller**, because the area is smaller.

Worked examples

Example 1: outstroke force
A piston has a diameter of 40 mm. The air pressure is 0.5 N mm^{-2}. Calculate the outstroke force.
r = 40 ÷ 2 = 20 mm
A = πr^2 = π × 20^2 = 1257 mm^2
F = P × A = 0.5 × 1257
F = 628 N

Quick check

1 A piston has a diameter of 60 mm. The air pressure is 0.5 N mm^{-2}. **Calculate** the outstroke force.

Space for working and answer

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Example 2: pressure needed
A force of 300 N is needed from a piston of diameter 30 mm. Calculate the air pressure needed.
$A = \pi r^2 = \pi \times 15^2 = 707\ mm^2$
$F = P \times A$, so $P = F \div A$
$P = 300 \div 707$
$P = 0.42\ N\ mm^{-2}$

Example 3: instroke force
The piston in Example 1 has a rod of diameter 10 mm. Calculate the instroke force.
rod area $= \pi \times 5^2 = 79\ mm^2$
effective area $= 1257 - 79 = 1178\ mm^2$
$F = P \times A = 0.5 \times 1178$
$F = 589\ N$

Controlling force

To get a **bigger** force: increase the air pressure, or use a piston with a bigger area.
A single-acting cylinder uses air only to outstroke: use the **whole** piston area. Its spring makes the instroke.

Watch out

Work out the **area** first. Never put the diameter straight into $F = P \times A$.
Keep pressure in **$N\ mm^{-2}$** with area in mm^2. Only use $N\ m^{-2}$ (Pa) if you change the area into m^2.

There is a mark for the **area** and a mark for putting the numbers into the formula. There is a mark for rearranging, if needed. The last mark is for the answer **with its unit**.

Check your understanding

1 **State** the formula that links force, pressure and area. **Give** the unit of each.

2 A piston has a diameter of 20 mm. **Calculate** its area.

Space for working and answer

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3 The air pressure is 0.4 N mm^{-2} and the piston area is 500 mm^2. **Calculate** the outstroke force.

Space for working and answer

4 **State** what effective area means.

5 **Explain** why the instroke force is smaller than the outstroke force.

Review Questions: now complete the Lesson Pn7 questions in your jotter.
Scan the QR code or go to **mrstewartphysics.co.uk/rq/pn7** · Tap **Show answer** to mark each one.

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Lesson Pn8: Describing and Designing Circuits

LI: To describe a circuit in QualScot words and design one to a specification.

SC: Pn8.1 I can **describe** a circuit step by step, starting from the words on the answer line. ☐

Pn8.2 I can **name** every valve and cylinder by its number or letter. ☐

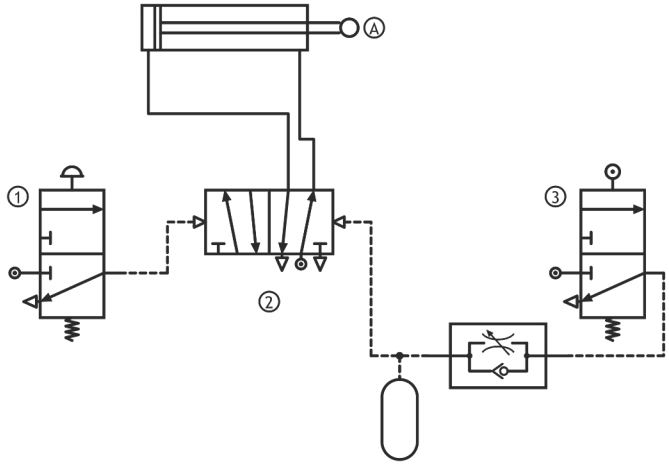
Pn8.3 I can **include** speed control and time delays in a description. ☐

Pn8.4 I can **design**, test and evaluate a circuit that meets a specification. ☐

How to describe a circuit

1. Start with the words on the answer line, e.g. "When valve 1 is actuated . . ."
2. Follow the air from valve to valve. Name every valve by its **number** and every cylinder by its letter.
3. Say what each valve does: it is **actuated**, it sends **pilot air**, or it **changes state**.
4. Say what each named piston does: it **outstrokes** or **instrokes**.
5. Say **slowly** if a uni-directional restrictor slows it, and say there is a **time delay** if there is a reservoir.
6. One mark is given for each step. A 5-mark question needs **five** steps.

Model answer



When valve 1 is actuated, pilot air changes the state of valve 2, so piston A **outstrokes**. At the end of its outstroke, piston A actuates valve 3. Pilot air from valve 3 fills the reservoir slowly through the uni-directional restrictor, giving a **time delay**. Then pilot air changes the state of valve 2 back, so piston A **instrokes**.

Words that get no marks

"It moves" gets no mark: say which piston, and whether it **outstrokes** or **instrokes**.
"UDR" on its own gets no mark: say what it **does** to the air.
"The valve turns on" gets no mark: say the valve is **actuated** or **changes state**.

Check your understanding

1 **State** the words a circuit description should start with.

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2 **State** what to name every valve and cylinder by.

---

3 **State** the two words to use for what a piston does.

---

4 **State** what to say if a circuit has a uni-directional restrictor and a reservoir.

---

5 A description question is worth 4 marks. **State** how many steps the answer needs.

---

6 **Explain** why "the valve turns on" gets no mark.

Review Questions: now complete the Lesson Pn8 questions in your jotter.
Scan the QR code or go to **mrstewartphysics.co.uk/rq/pn8** · Tap **Show answer** to mark each one.

Build It: the Bottle Crusher

A prototype system crushes plastic bottles. For safety, two buttons must be pressed before the plastic is crushed by a pneumatic cylinder. After a set time, the piston must instroke automatically (2017 Specimen Question Paper Section 2 Question 16).

The following specification has been written:

A Piston A outstrokes only when push-button valves 1 and 2 are both actuated.
B Piston A outstrokes slowly and smoothly.
C After an adjustable time delay, piston A instrokes automatically.

Pn8a Draw the circuit in the box, using the symbols on the data sheet (page 2). Label every valve with a number.

Pn8b Write a test plan: three tests and the expected result of each. After you have simulated or built it, record each test result. Say what the piston did.

Planned test	Expected result	Test result

Pn8c Simulate the circuit, then construct it on the kit.

1. Open NoPressureSim (QR code). **Build** the circuit, as on page 8. Show the main air supply on every valve.
2. Predict what happens. Test it. Was your prediction right?
3. Air off. **Build** your circuit on the kit.
4. Air on. Carry out your three tests. **Record** each test result.

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Pn8d Evaluate your solution. For each specification point, state if it was met and explain why, using your test results.

Specification point	Met? Yes/No	Explanation
A		
B		
C		

Pn8e Large bottles are not fully crushed before the piston instrokes. **Describe** a modification that would solve this. Justify it.

Modification	
Justification	

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Knowledge Organiser

Key word	What it means
main air	Air that operates a cylinder (makes the piston outstroke or instroke).
pilot air	Air used to actuate a valve. Drawn as a dashed line.
exhaust	Where used air leaves the circuit to the atmosphere.
actuated	A valve is actuated when it is operated (pressed, moved, or switched by air or electricity).
outstroke / instroke	The piston moves out of / back into the cylinder.
3/2 valve	3 ports and 2 states. Operates a single-acting cylinder or sends pilot air.
5/2 valve	5 ports and 2 states. Operates a double-acting cylinder. Stays in its state until pilot air reaches the other end.
reciprocating	Outstroke then instroke, over and over.
uni-directional restrictor	Slows the air in one direction only. Slow the exhaust air for slow, smooth motion.
time delay	A uni-directional restrictor fills a reservoir slowly before a valve changes state.
AND control	Two valves in series: both must be actuated.
OR control	Two valves and a shuttle valve: either one works.
$F = P \times A$	Force (N) = pressure (N mm^{-2}) × area (mm^2). $A = \pi r^2$.
effective area	Piston area − rod area. Used for the instroke force.

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